

DS3000 Canadian Housing Burden Analysis

Group 11

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Word Count: 3286

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Overview

This report is focused on analyzing Canadian households and their household disposable income spent on shelter expenditures, with data provided by Environics Analytics. The target variable, the HBR (housing burden ratio), was defined as shelter expenditure over disposable income. Both unsupervised and supervised learning were performed to better understand how different Canadian households were affected.

Data Prep

Since this was a large dataset split over multiple files, it was loaded with lazyframes and joined on both CODE and GEO columns. This produced a table with 950,157 rows and 448 columns, which was then filtered to 57,936 rows and 448 columns to only include rows with PRCDDA in the GEO column. The next step was creating the HBR column to hold the target variable as defined above. During this step, it was found that certain rows had 0 for disposable income, meaning their HBR value would be infinity. 1079 rows (<2%) would have been dropped, so dropping them was unlikely to introduce bias. The next step was to remove leakage by dropping variables that had any correlation with HBR, like any variable starting with HSSH, along with some aggregates, leaving 393 out of the 448 original columns. Next, the final tableau was constructed, which included the demo data merged on the CODE column to create a table with 1128 columns. 0-variance columns were also checked, and surprisingly, there were none. Another check was to look for any missing values, and none were found, meaning no imputation or deletion was necessary for the data. If there were missing values, we would have used median imputation to handle it, as our graph is skewed and contains outliers. Lastly, the CODE and GEO columns were removed before export for use in the rest of the analysis. With this, the final table contained 56,857 rows and 1125 columns.

EDA

The merged table contained 56,857 rows and 1125 columns, including the target variable, which gives us a final feature count of 1124 columns after all the removals and leakage and was used for modelling. Non-numeric and zero-variance columns, such as HSSH-prefixed variables, were dropped, as the model should identify which household characteristics were associated with housing burden. Leaving these values could cause the model to memorize or have an overly confident R^2 score because HSSH-prefixed variables are added to the numerator. Afterwards, an 80/20 split of the newly processed data into a training and testing set was done, resulting in 45,485 training points and

11,372 for testing. The training set was then scaled using a standard scaler and applied to the test set to ensure no leakage from the test set.

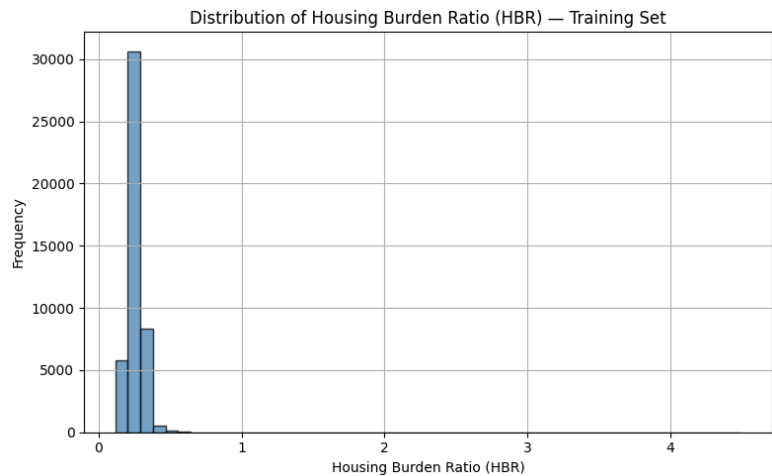


Figure 2.1

A histogram was used to view the distribution of the housing burden ratio before calculating the mean, median, standard deviation, and skewness (Fig. 2.1). By examining the distribution of the target variable from the graph and the skewness value of 15.4075, it was evident that the graph was heavily right-skewed. It was also evident that most values fell between 0.20 and 0.35, near the mean of 0.257 and median of 0.251, with some extreme outliers like the 4.49 right tail. The mean and median being almost identical revealed how the distribution was fairly symmetric. However, because of this skewness, it motivated the Pearson Transformation in part 3.

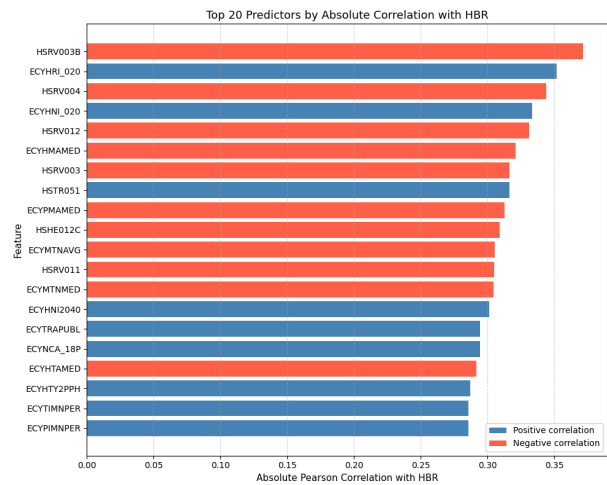


Figure 2.2a

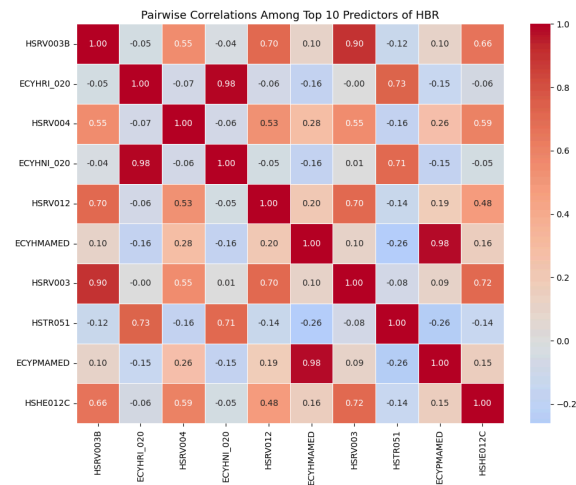


Figure 2.2b

A bar chart was used to display the correlation certain features had with the target variable (Fig. 2.2a). This observation showed how the top 20 correlated variables fell between 0.28 and 0.37, meaning that no single feature had a significant impact on housing burden and that a combination of multiple features could be the result of such effects on housing burden. The strongest negatively correlated feature was **"HSRV003B"** (Purchase of snowmobiles), achieving -0.3713, while the strongest positively correlated feature was **"ECYHRI_020"** (Household Income \$0 To \$19,999). These observations seemed to align with intuition, where wealthier households, with less housing burden, could afford luxury goods like snowmobiles, whereas impoverished households earning less than \$20,000 were more prone to housing burden.

The heatmap (Fig. 2.2b) revealed that some of the top variables were heavily correlated with each other. For example, **"ECYPMAMED"** (Median Age Of Male Population) and **"ECYHMAMED"** (Median Age Of Household Population Male) achieved a correlation score of 0.98, effectively making both variables the same. The same could be seen with **"ECYHNI_020"** (Household Income \$0 To \$19,999 (Current Year \$)) and **"ECYHRI_020"** (Household Income \$0 To \$19,999 (Constant Year 2020 \$)), also achieving a correlation score of 0.98. With this in mind, having a regularization technique like LASSO or Ridge could be extremely effective in preventing overfitting and underfitting.

K-Means

Table 3: Inertia and Silhouette Scores for Each K Value from 2 to 12

K	Inertia	Silhouette Score
2	38480999.7	0.7765
3	32687896.5	0.5392
4	30307733.1	0.3997
5	28876872.1	0.2644
6	27924032.7	0.1760
7	27338074.1	0.0798
8	26810994.8	0.0822
9	26197965.5	0.0940
10	25823387.7	0.1108
11	25383279.6	0.0477
12	25058384.1	0.0223

KMeans Model Selection

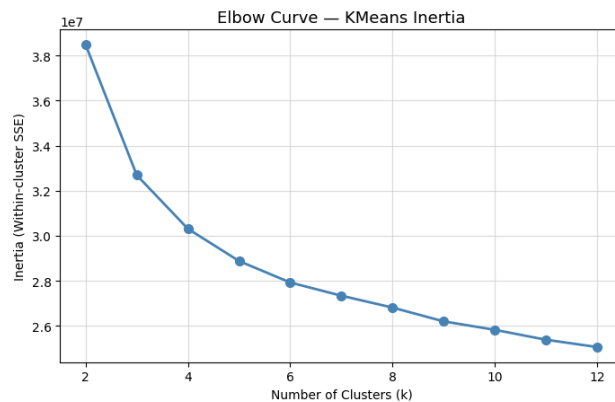


Figure 3.1a

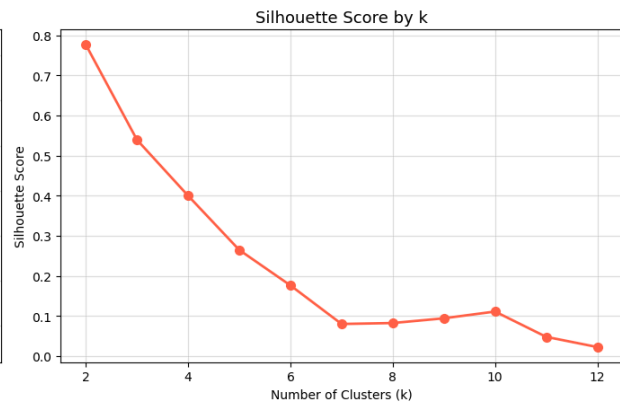


Figure 3.1b

We tested different k values on a training set consisting of 45,485 observations and 1124 features from 2 to 12. Both inertia and silhouette score were recorded, and it was discovered that k = 2 (0.7765) was the highest silhouette score. Note that only a subset of 5000 samples was used when calculating the silhouette score. Although this may have caused some variance, the overall trend across k remained reliable while increasing efficiency. Around k = 3, there was also a visible bend in the elbow plot, before slowing down (Fig. 3.1a). The two methods partially disagreed in the number of clusters; however, the team decided to use the elbow plot to determine the optimal k of 3. This was because the data type, households, was more likely to be better represented by multiple categories rather than a single binary value of “high” or “low”

burden. The preprocessing done earlier also likely had an influence on how the optimal k was determined. As the entire dataset was used to determine the scores, the distance between every point was similar, resulting in a bias towards lower dimensions.

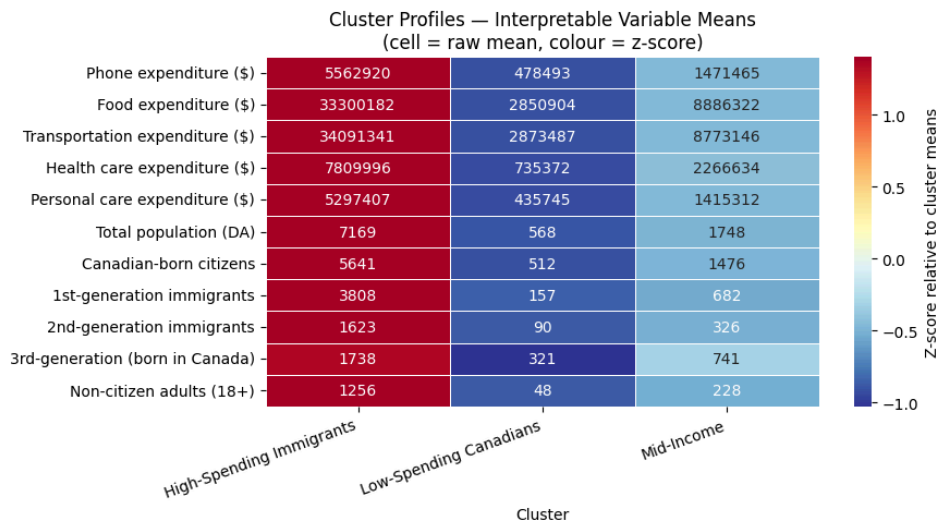


Figure 3.2

Because k=3 was chosen, there were three clusters classified as Small, Mid and Large DAs (Dissemination Areas) (Fig. 3.2) with 40,895, 4357, and 233 samples, respectively. The first cluster contained a population of 568, mostly Canadian-born and low spending, which led to the label “Low-Spending Canadians”. The second cluster, containing a population of 1748, was mostly moderate across the board, which led to the label “Mid-Income”. Lastly, the third cluster contained a population of 7169, were on the higher end of most statistics, like spending as well as being mostly immigrants, which led to the label “High-Spending Immigrants”. From this chart, it could be seen that all expenditure and population variables were larger by 6-7x in Cluster 1. The scaling was also not uniform; to be specific, 1st-generation immigrants scaled at 11x.

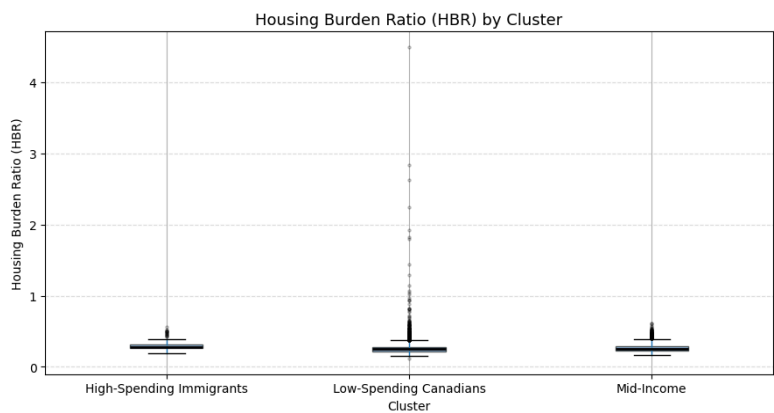


Figure 3.3

Plotting these clusters in the boxplot (Fig. 3.3) showed that all three groups had similar median HBR of around 0.25. However, the Low-Spending Canadians had a much higher outlier count, with some reaching past 4 HBR. This indicated that the most severely housing-burdened households were in small, lower-spending areas rather than more urban areas. The high-spending immigrants were also slightly more compact, with fewer outliers and more households closer to the mean. Overall, these clusters mostly represented size rather than HBR.

PCA

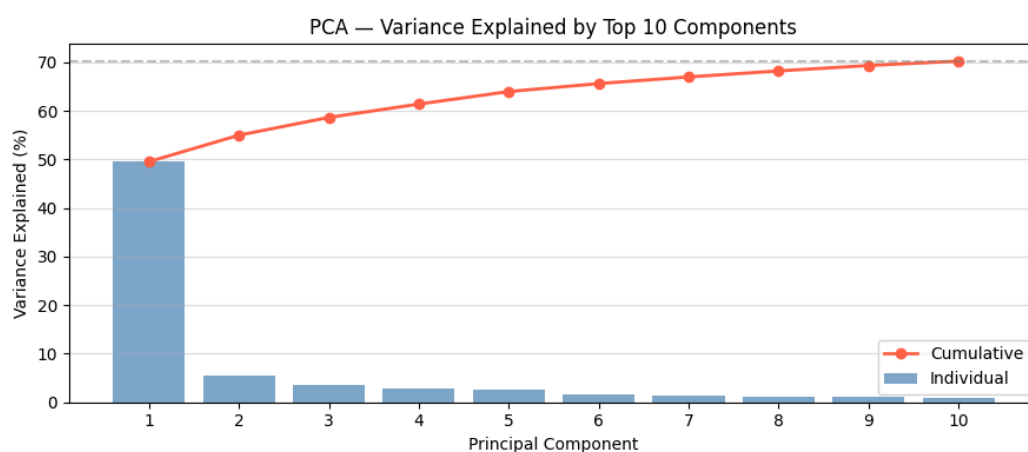


Figure 4.1

PCA was fit on the training set. PC1 captured the majority of the variance (~50%), while PC2 captured only ~5%, and the rest followed around that size. This steep drop-off from PC1 to PC2 explained how the dataset could be dominated by a single factor with several secondary dimensions capturing smaller parts.

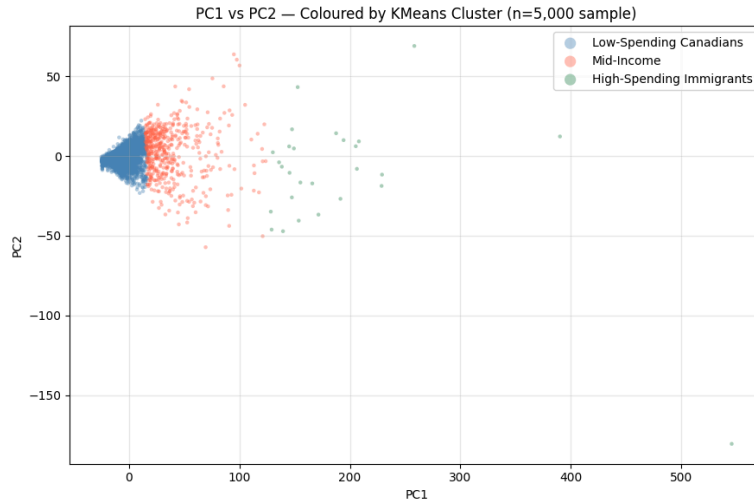


Figure 4.2

Comparing PC1 and PC2 on 5k samples, Low-Spending Canadians formed a tight cluster near the origin, while the High-Spending Immigrants spread along the PC1 axis. As for the cluster assumption, the data did not support it because of the lack of space between classes and the heavy imbalance due to the majority of points being clustered near the origin.

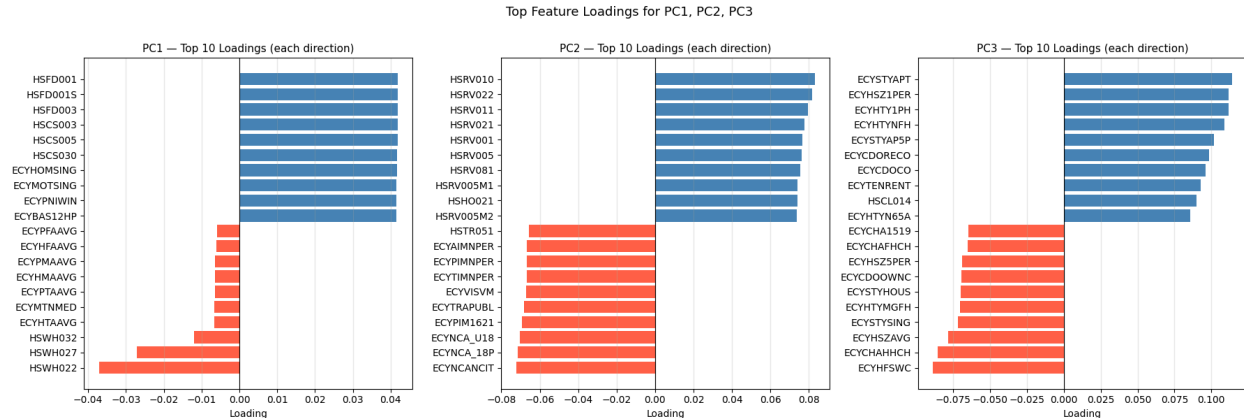


Figure 4.3

For PC1, the positive loadings identified were expenditure variables like food and telecom, while the top negative loadings were all real estate-related. This component separated households that spend on everyday goods from those that sell property.

PC2 represented Rental Housing Supply versus Recent Immigration. Their positive loaders consisted of all “HSRV*” features, representing rented dwelling variables. Their negative loaders consisted of recent immigration, public transit, and

non-citizens variables. This component separated areas with owner-occupied housing from immigrant neighbourhoods that use public transit.

PC3 represented Urban Apartments versus Family Homeowners. Their positive loaders consisted of apartment-style buildings and renting households/apartments. Their negative loaders consisted of multiple people in a household, such as having children in their home. This component separated a dense urban single occupant from suburban family homes.

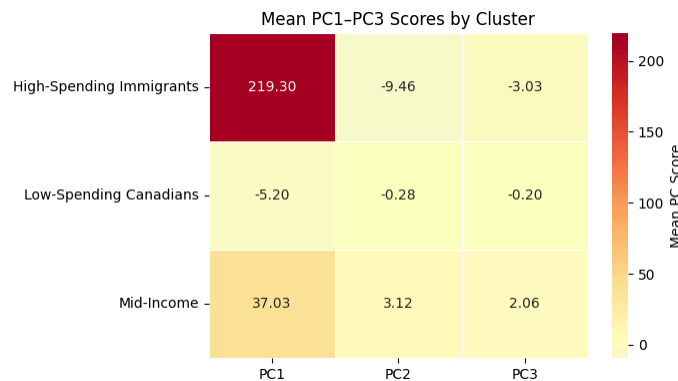


Figure 4.4

Computing mean scores by clusters, PC1 immediately dominates cluster differentiation. PC1 was how big the neighbourhood was, and High-Spending Immigrants scored a +219.30, almost ~7x bigger than Mid-Income's PC1 score, putting them far in the large Dissemination Area. PC2 and PC3 showed lower differences with High-Spending Immigrants scoring negative on both, Mid-Income people scoring positive on both, and Low-Spending Canadians also being negative for both.

From these mean PCA scores, cluster names could be altered into a more human-readable format:

High-Spending Immigrants → Dense Immigrant Urban DAs. They had a large PC1 value (big neighbourhood) with negative PC2 and PC3 values, so this cluster was more transit-dependent and lived in apartments/single houses.

Mid-Income → Medium Scaled Established Suburban Family DAs. They had a moderate PC1 score (medium-sized) with positive PC2 and PC3 scores, indicating families with children in established houses.

Low-Spending Canadians → Small Neutral Canadian DAs. They achieved scores close to 0 for all PCs, indicating they were slightly below average in size but neutral in everything else.

The main difference between the human-readable cluster names and the previous ones was that the PCA names described what kind of place it was, rather than what it was. K-Means names represented the spending levels and who lived there, with names like “Low-Spending Canadians”, which showed that they were Canadians and spent less. The PCA names like “Medium Scaled Established Suburban Family DAs” showed these as more suburban medium-sized neighbourhoods with more owner-occupied housing with families that had children, not just “Mid-Income”. These new names used the PCA means to help characterize the clusters and differentiate them by revealing details like size and housing.

UMAP

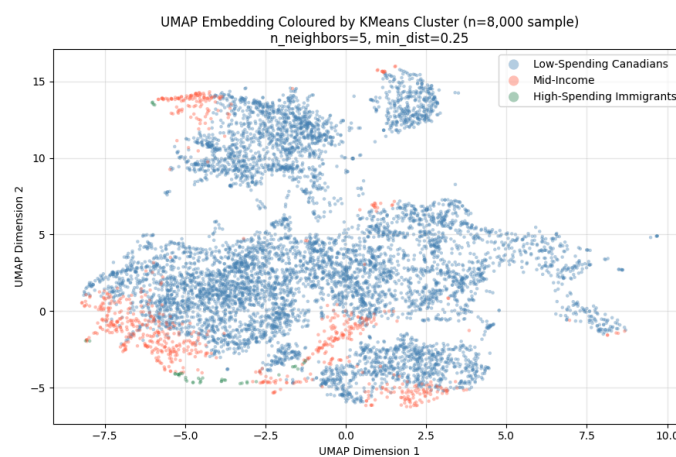


Figure 5.1

UMAP was applied to the training set to reduce the data into two dimensions. To figure out the optimal parameters for the best output, different `n_neighbour` values were tested: 5, 15, 30, and 50; along with `min_dist` values: 0.0, 0.1, and 0.25. These 12 combinations were tested on an 8000-point subsample. This resulted in finding a silhouette score of 0.0951 with the parameters of `n_neighbours` being 5 and `min_dist` being 0.25 (Fig. 5.1).

In the Scatterplot, a partial separation could be seen between the three clusters with some overlap. However, K-means and UMAP captured different structures in the data, as the K-means groupings did not align with UMAP’s low-dimensional reduction.

Whether UMAP was better or worse than PCA depended on the goal. PCA provided a cleaner linear separation along PC1, where the three clusters were distinguishable by their aggregate scores. UMAP would be better for more nonlinear structures, and while it may have worked well on the low/high spending clusters, the mid cluster lacked coherence in UMAP space, which suggested that K-means may have forced a linear boundary through a continuous gradient.

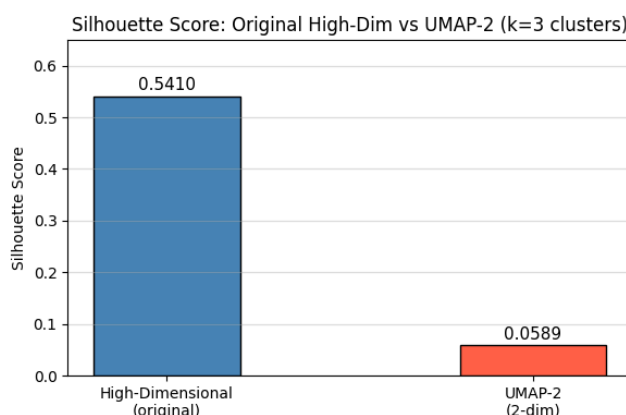


Figure 5.2

After computing the silhouette score in 2D UMAP space and comparing it with the original high-dimensional score (Fig. 5.2), the high score was driven mostly by low-spending Canadians, who were the points packed closely in the high-dimensional space. On the other hand, the silhouette scores dropped from 0.5410 to 0.0589 in UMAP, which could be attributed to compressing 1124 dimensions to 2, and UMAP prioritizing local neighbourhoods. This graph also showed how the K-Means partition ($k=3$) was only partially real since the low-spending cluster contained multiple hidden sub-markets. In short, the high-spending immigrants group is the most informative as it forms a cohesive group in both UMAP and PCA space and achieves the highest silhouette scores.

ElasticNet

Since the data was heavily skewed, the distribution of the target variables (y_{train}) was normalized, allowing it to work with a linear model like ElasticNet.

Grid search was performed to find the optimal parameters for the ElasticNet model. To save time, the team opted to do parameter estimation with a single-layer neural net with an ElasticNet loss function (MSE + L1 + L2 penalties). Adam was chosen as the optimizer along with full-batch passes to remove gradient noise, producing stable CV R^2 results comparable to the original ElasticNet GridSearchCV. A

known limitation was that Adam could not force coefficients to exactly zero, so this method was merely used for parameter approximation and selection. The final model was trained as an ElasticNet model using the optimal parameters identified with the approximation.

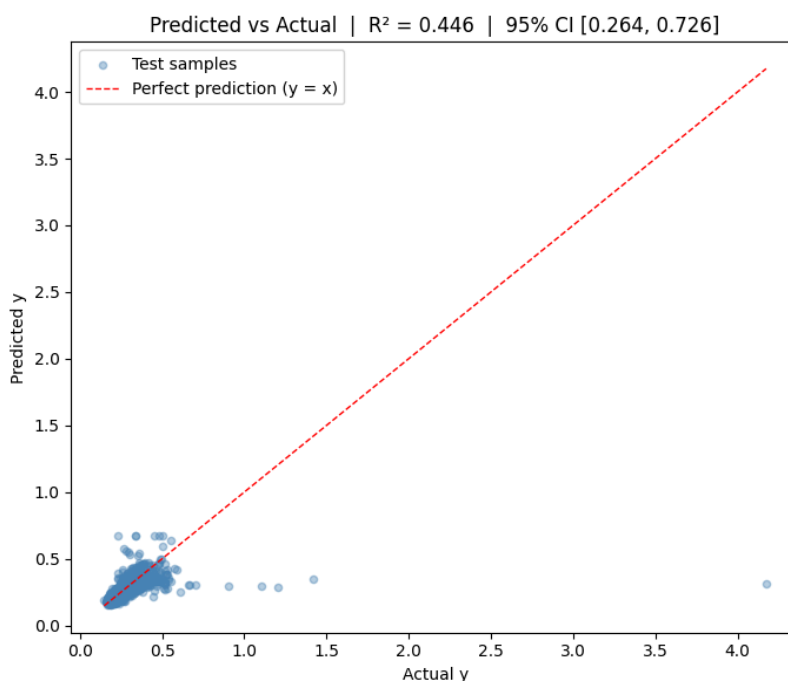


Figure 6.1

After running the parameter search, the best parameters found were an alpha of 0.001 and an L1 ratio of 0. The low alpha value and L1 ratio suggested that the model was weaker at feature selection and that all variables played a non-zero role in the model. Because the L1 ratio was 0, this meant that the best model was essentially a ridge regression model, so a ridge model was trained rather than wasting computation time on ElasticNet coordinate descent.

The model reported a test R^2 score of 0.440 with 95% confidence between [0.259, 0.718], meaning the model was able to capture a decent amount of structure. An MSE of 0.0022 was also found, which meant the model had a low amount of prediction error on the test set (average error of $\sqrt{0.0022}$ standard deviations). Looking at the scatterplot (Fig. 6.1), many of the points were near the red “perfect” line, with a few clear outliers.

The two most populated clusters, low-spending Canadians and mid-income Canadians, both had a reasonable R^2 score (0.4433 and 0.6538, respectively), with better performance for mid-income Canadians. For high-spending immigrants, the

model had an R^2 score of -2.5642, meaning significantly worse performance than predicting the mean. The model, being trained on Canadian data, likely did not fit well for immigrants. Also, wealthier households had more choices for housing, often choosing based on values and preferences, whereas poorer households did not have the privilege of choice and had similar forced spending habits.

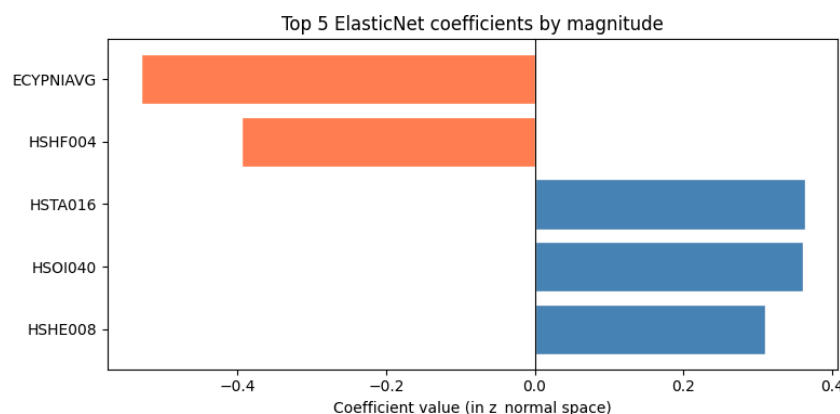


Figure 6.2

The top five coefficients (Fig. 6.2) by magnitude are average income, rugs/mats, recreational cannabis, contract work for repair/maintenance, and portable AC and humidifiers. Average income made sense because there was a minimum amount that households must spend on shelter, regardless of income. Logically, rugs proxy house size, which inversely correlates with burden. Cannabis usage also often increases in lower-income areas. Finally, it was possible that lower-income housing requires more repairs (due to poorer conditions) and does not come with functional HVAC systems, requiring portable substitutes.

XGBoost

Considering the large dataset, the team opted to use the 'hist' tree method for the XGBRegressor, as it sped up computation time without sacrificing significantly on performance. An estimator count between 100 and 600 was chosen, and the max depth was up to 10, which allowed it to capture complex patterns. The learning rate was also set between 0.01 and 0.21, and the subsample and colsample_bytree values were both 0.6-1.0. Although that normally risks overfitting, the extremely large dataset protected against overfitting well. To catch correlations/interdependencies, the trees were configured to see more features rather than being restricted to smaller subsets.

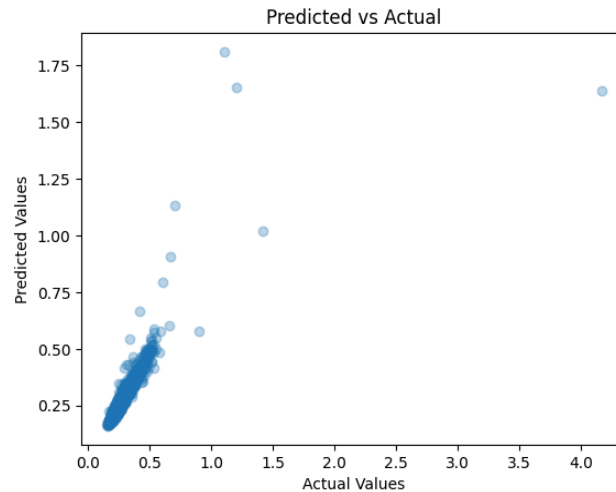


Figure 7.1

After running the model, it was clear that XGBoost's performance was significantly better than ElasticNet, with an R^2 score of 0.80 (95% confidence between [0.6945, 0.9413]) and a test MSE of 0.0008. The reasoning for XGBoost's superiority is that the dataset contained a lot of interdependent and redundant features. Each category had many extremely similar features, and many were dependent on each other. This made linear models like ElasticNet weak because it treated variables as independent and useful. XGBoost was capable of finding dependencies, making it have much higher predictive capabilities using the dataset. The team recommends that any decision-makers use XGBoost (and other non-linear models) over ElasticNet for similar studies.

SHAP

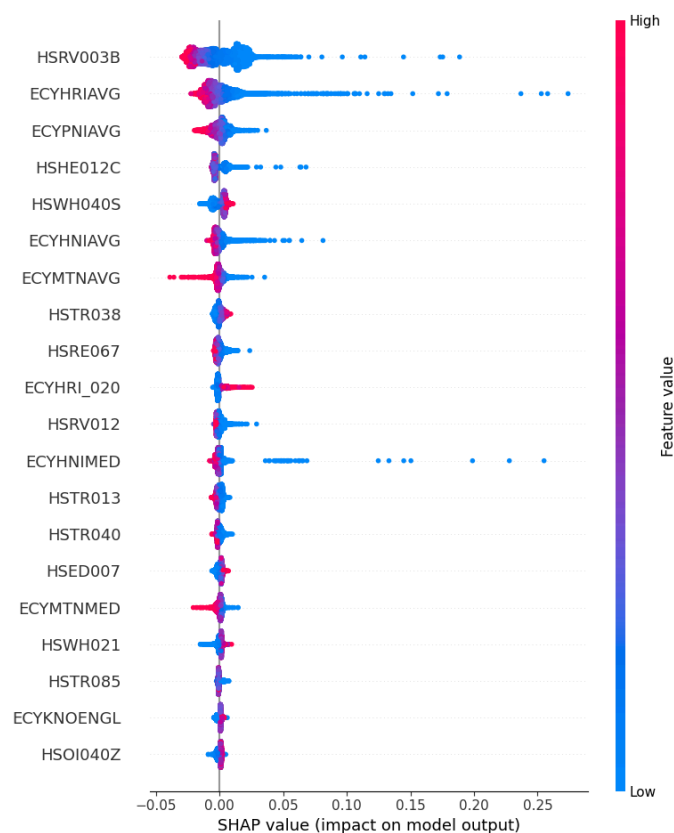


Figure 8.1

The beeswarm plot ranked the coefficients by importance. The top three variables each had their high values on the negative side, suggesting negative correlations with the target. However, many features contained varying data points that pulled the prediction in either direction, suggesting features may also have been influenced by other factors, like interdependence and correlations.

Top 10 Features

1. Purchase of Snowmobiles
2. Average Household Income for the Year 2020
3. Average Income
4. Parts and Accessories for Gardening Tools
5. Net purchase price of owned residences
6. Average Household Income for Current Year
7. Average Maintainer Age
8. Parking costs
9. Television and satellite radio services
10. Household Income less than \$20k in Year 2020

Surprisingly, snowmobiles were ranked as the most important variable. This was likely due to the tree finding a hidden but real correlation between purchasing this luxury item and shelter burdens. Trees only find predictive patterns and NOT causal relationships. Similarly, only those who had houses with backyards would likely justify purchasing gardening tools.

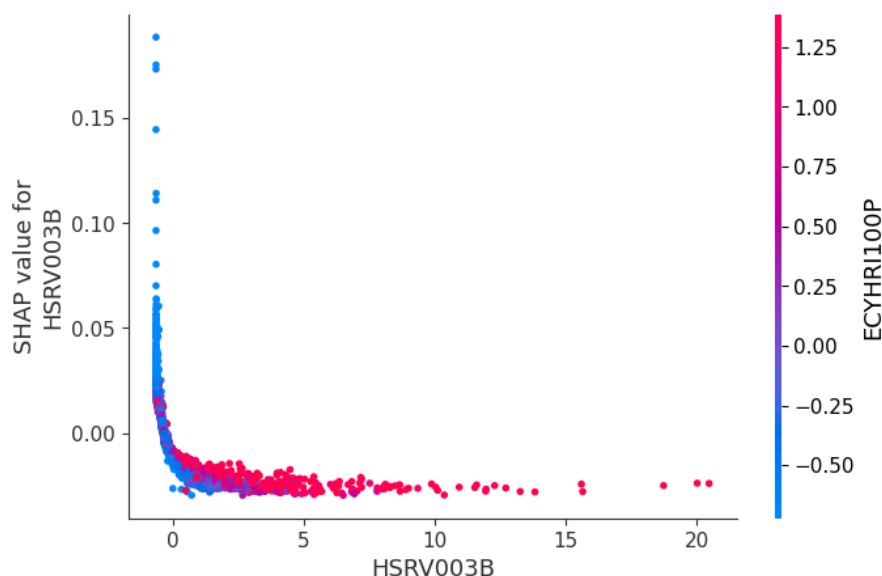


Figure 8.2

Because of how surprising the top coefficient was, a dependence plot was run on it (Fig. 8.2). It was compared against household income over \$100,000, a reasonable income to justify purchasing snowmobiles. From the plot, low values of income (blue dots) were correlated with higher SHAP values, hence higher predictions of shelter burden. Meanwhile, values with low SHAP values were typically associated with high income (red dots). This suggested that snowmobile purchase may indeed be a proxy for income.

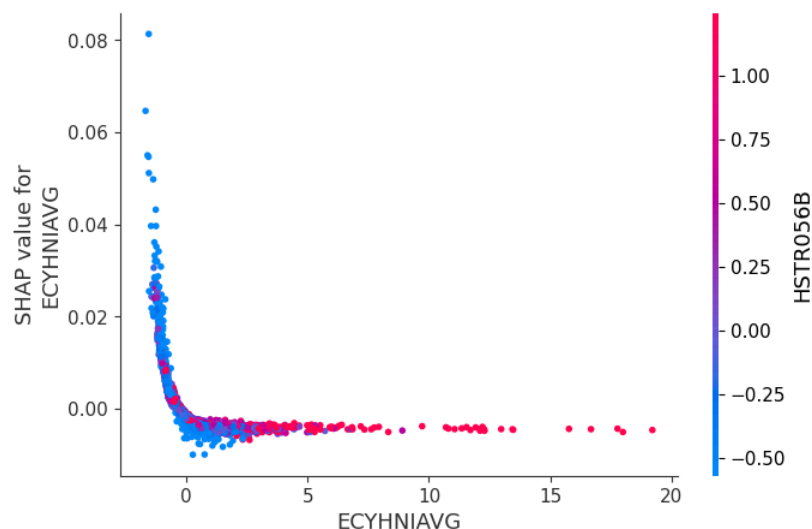


Figure 8.3

A test for Average Household Income for the current year was also run (Fig. 8.3). Unsurprisingly, the plot was similar to the previous one, suggesting that those who had higher income in today's year had less shelter burden.

This problem was non-linear because its features were heavily correlated and likely had high levels of dependencies between them, which was something a linear model could not account for. The top variables, except average income, differed from those of ElasticNet, but all variables were likely related to income in some way, so it could not be said that one model's top coefficients were objectively better.

Looking at the median HBR across all 3 groups in Figure 3.3, they were almost identical; therefore, these clusters did not meaningfully identify HBR. Low-spending Canadians had many more extreme outliers, while the high-spending immigrants were clustered more around the median. Evidently, they were most likely separated by population composition rather than HBR, as shown by PC1 being the dominant component, while the supervised model chose the dominant features to be income and spending.

Conclusion

Overall, this report analyzed various factors which caused housing burden across Canadian households by combining both unsupervised and supervised learning techniques. The clustering analysis clearly indicated distinct household groups with meaningful differences in income, composition and several other features; however,

these distinct groups showed little variation in HBR. While PCA was able to capture most of the linear structure, UMAP revealed additional non-linear ones, revealing that household behaviour was inherently complicated. Given the non-linear nature of the problem, XGBoost substantially outperformed Elasticnet. A known limitation is that associations were much easier to prove compared to causation, a constraint inherent to cross-sectional data.